Tapas Barman

MLOps Engineer | AI Engineer | Data Scientist

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Professional Summary

Results-driven MLOps & AI Engineer pursuing M.Sc. Data Science at IIIT Lucknow (SGPA 8.85), with production experience at IIT Bombay and Aaizel Technologies. Specialises in end-to-end ML pipeline automation, LLM-powered agentic systems, and secure CI/CD for model governance. Demonstrated business impact: reduced DSM financial penalties by **95%** (24.8L → 1.3L) via physics-informed ML forecasting, projecting **1.41 Cr** in annual savings. Seeking a full-time role in MLOps, AI Engineering, or Applied Data Science.

Experience

MLOps Engineer Intern Dec 2025 – Present
Aaizel International Technologies Pvt. Ltd. *On-Site*

- Architected a production-grade ETL and ML inference pipeline for **Climate Forte** (75 MW wind platform), orchestrating multi-step DAGs with **Apache Airflow** and persisting outputs to **MinIO/S3-compatible** storage.
- Implemented full ML lifecycle management — **MLflow** experiment tracking, **DVC** data versioning, **Evidently AI** drift monitoring — alongside **Jenkins** CI/CD, enabling automated retraining with zero-downtime champion–challenger model promotion.
- Built ML backend for **VidyaDristi (OSINT)** leveraging **NER**, **LLM embeddings**, **FAISS vector search**, and density-based clustering to surface actionable intelligence from unstructured data sources.
- Secured the build chain using **SonarQube** (SAST), **Trivy** (container scanning), **Gitleaks** (secret detection), and **Nexus** artifact management — achieving policy-compliant deployments across all model versions.

AI Software Research Intern Jul 2025 – Oct 2025
Indian Institute of Technology Bombay (IIT Bombay) *Remote*

- Contributed to **Kalanjiyam** — IIT Bombay's digital preservation platform for ancient Siddha literature — by engineering two production OCR sub-systems for Tamil and Sanskrit manuscript digitisation.
- **(Leap OCR)** Replaced legacy **Tesseract** engine with **GPU-accelerated EasyOCR**, achieving **3× throughput improvement** and **18% reduction in Character Error Rate (CER)** on Indic-script documents.
- **(Bhaasha OCR)** Designed table detection and structural reconstruction pipeline using **YOLO + Transformer-based models**, improving table-level **TEDS accuracy by 22%**; outputs integrated via **REST API** into the Kalanjiyam system.
- Evaluated model quality using **CER, WER, BLEU, and TEDS** metrics; delivered validated pipelines integrated with the platform's REST API layer for end-to-end document processing.

Projects

Wind DSM Optimizer — Physics-Informed ML Forecasting Platform
Python | XGBoost, Scikit-learn, MLflow | FastAPI, Apache Airflow, Docker | NOAA GFS, Pandas **GitHub**

- Built a **physics-informed ML forecasting pipeline** for a 75 MW wind plant generating **96-block (15-min) day-ahead** generation schedules, improving forecast accuracy by **71.2%** over baseline.
- Reduced DSM penalty liability from **24.8L to 1.3L** per cycle, projecting **annual savings of 1.41 Cr** — a direct, quantified business outcome delivered via ML.
- Automated the full inference workflow via **Airflow DAGs** ingesting live **NOAA GFS** NWP data; predictions served through a **FastAPI REST endpoint** with **MLflow champion–challenger** model versioning.

Production MLOps Platform — Secure CI/CD & Model Governance
Python | MLflow, DVC | Jenkins, Docker, Evidently AI, SonarQube, Trivy, Gitleaks, Nexus **GitHub**

- Engineered an end-to-end MLOps platform covering **data versioning (DVC)**, **experiment tracking (MLflow)**, containerised inference **(Docker)**, and automated model promotion from development to production.
- Embedded production-grade quality gates into **Jenkins** pipelines: data drift **(Evidently AI)**, algorithmic bias checks, static analysis **(SonarQube)**, container vulnerability scanning **(Trivy)**, and secret detection **(Gitleaks)**.

AI Trend Notifier — Autonomous Agentic Monitoring System
Python | LangGraph, LangChain | DistilBERT, Groq LLaMA, Tavily API | HuggingFace, Streamlit, SQLite **GitHub**

- Designed a **multi-agent pipeline** using **LangGraph** to autonomously ingest AI trend signals from Twitter and Reddit, apply **DistilBERT-based sentiment classification**, enrich context via **Tavily web search**, and generate LLM-synthesised summaries using **Groq LLaMA**.
- Developed a production-ready application with **Streamlit** frontend and **FastAPI**-style backend supporting real-time trend visualisation, subscriber management, and automated **email newsletter delivery** (SMTP + SQLite).

Technical Skills

Languages & Systems: Python, SQL, Bash, Linux, Git, Docker, Docker Compose
MLOps & Data Engineering: MLflow, DVC, Apache Airflow, Apache Kafka, Evidently AI, MinIO/S3, Snowflake, FastAPI
CI/CD & DevSecOps: Jenkins, GitHub Actions, SonarQube, Trivy, Gitleaks, Nexus
Generative AI & LLMs: LangChain, LangGraph, RAG Pipelines, FAISS, Groq LLaMA, Tavily API, HuggingFace Transformers
Model Training & Optimisation: PyTorch, Scikit-learn, XGBoost, LoRA, QLoRA, Unsloth, Quantisation (bitsandbytes), Accelerate
Data & Visualisation Stack: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, SQLite

Education

Indian Institute of Information Technology (IIIT), Lucknow — 2024 – 2026
M.Sc. Data Science ***SGPA: 8.85 / 10*** — *Lucknow, UP*

Acharya Prafulla Chandra Roy Government College (University of North Bengal) — 2020 – 2023
B.Sc. Physics ***CGPA: 8.23 / 10*** — *Siliguri, WB*

Achievements

- **IIT-JAM 2024** — Secured **All India Rank (AIR) 1200** in Physics out of $\sim$18,000 candidates; qualifying for IIIT Lucknow M.Sc. admission.
- **LeetCode** — Solved **200+ algorithmic problems** spanning arrays, dynamic programming, graphs, and system design.
- **Kaggle** — Active practitioner with public notebooks in NLP and tabular modelling; competitions in ML classification tasks.

Certifications

- **Google Data Analytics Professional Certificate** — Coursera — *Aug 2024*
- **IBM Data Science Professional Certificate** — Coursera — *Jun 2024*
- **Big Data Analysis with GCP** — Udemy — *May 2024*